

Quantum Computation and Information

Homework Set 3

1. Given two vectors $x, y \in \mathbb{R}^n$, using amplitude encoding we can encode them as quantum states

$$|x\rangle = \sum_{k=1}^n \frac{x_k}{\|x\|} |k\rangle, \quad |y\rangle = \sum_{k=1}^n \frac{y_k}{\|y\|} |k\rangle,$$

where $\|\cdot\|$ denotes Euclidean norm and $\{|1\rangle, |2\rangle, \dots, |n\rangle\}$ are orthonormal quantum states.

(a). Show that the inner product $x^T y = \|x\| \|y\| \langle x|y\rangle$.

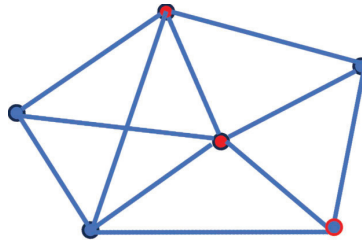
(b). The outer product xy^T is an n -by- n matrix. How to encode xy^T quantumly using $|x\rangle$ and $|y\rangle$?

(c). Suppose the answer to (b) is A . Show that the inner product $x^T y = \text{tr } A$.

(d). Design a quantum algorithm to determine the sign of $x^T y$, i.e., if it is positive or negative.

2. Given N vectors $x_1, x_2, \dots, x_N \in \mathbb{R}^n$. Define an N -by- N matrix K with entry $K_{ij} = x_i^T x_j$. Matrix K can be encoded as a quantum state described by density matrix $\rho = \sum_{i=1}^N \sum_{j=1}^N K_{ij} |i\rangle \langle j|$, where $\{|1\rangle, |2\rangle, \dots, |N\rangle\}$ are orthonormal quantum states. Explain in detail how you would prepare quantum state ρ .

3. Given the following graph and assume that the weight of each edge is 1. Find its max-cut using variational quantum algorithm. You can either implement it using Qiskit or TensorFlow Quantum (if you have worked with it before or prefer).



[For the submission of this problem, please email me the notebook.]